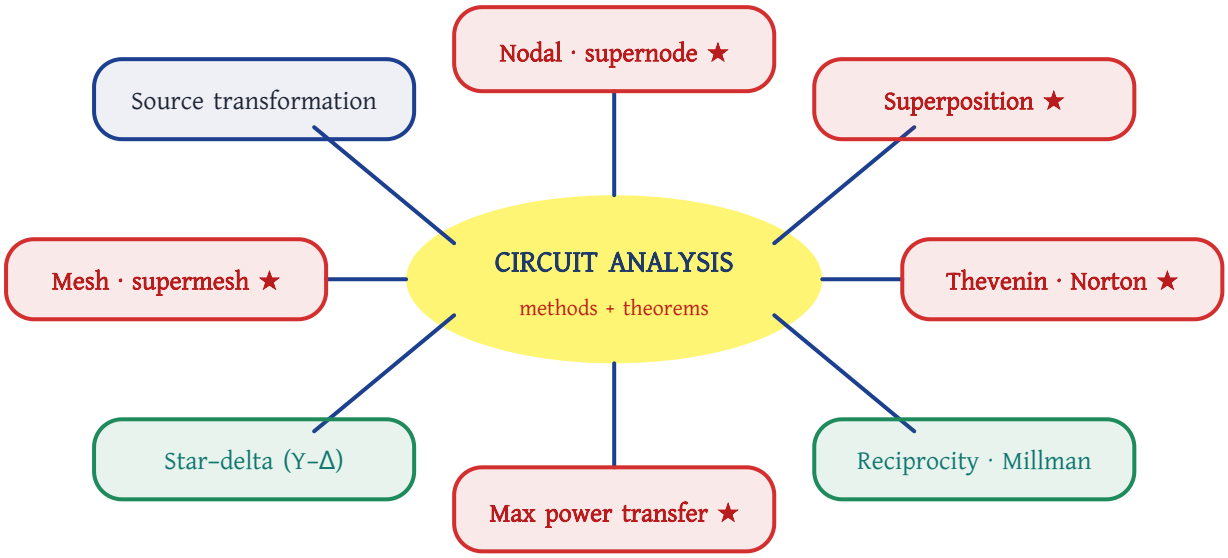
Circuit Analysis & Theorems

Nodal · Mesh · Superposition · Thevenin/Norton · Max power · Y–Δ: exam-ready in 11 sheets



↑ the whole lesson on one page: red = high-yield, green = applied / extra

Syllabus coverage

"Simple resistive circuits: series and parallel circuits, voltage and current division, **source transformation**. Methods of analysis: **nodal and mesh analysis**. Circuit theorems: **Thevenin's, Norton's and superposition theorems**, **maximum power transfer** and reciprocity theorem". Ohm's law, KCL/KVL, series-parallel and the dividers are in the **Basic Laws** notes (lesson 1); phasor (AC) versions of these methods are in lesson 3.

High-yield list

① meshes = b − n + 1
② V-source off → short, I-source off → open
③ $V_{th} = V_{oc}$, $I_N = I_{sc}$
④ $R_L = R_{th}$, $P = V_{th}^2/4R_{th}$, η = 50%
⑤ no superposition of power
⑥ Δ→Y: product ÷ sum

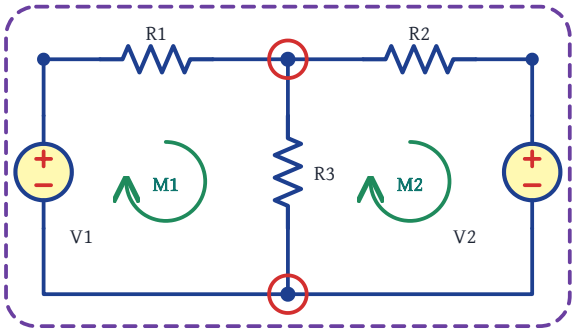
Legend: **yellow** = key term · **red circle** = must-memorise fact · **purple box** = memory trick · **red box** = exam trap · **green** = worked example

Contents

✓ Network terms, choosing a method … p. 2
✓ Nodal analysis ..... p. 3
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1 The Language of Network Analysis

Term	Meaning
Node	2 or more elements join
Essential node	3 or more elements join
Branch	one element (or series string)
Loop	closed path, no node twice
Mesh	loop with no loop inside it
Planar	drawable flat, no crossing



dashed = 3rd loop (not a mesh)

red rings = essential nodes

Counting equations: b branches, n nodes: nodal needs **n − 1** equations · mesh needs **b − n + 1** equations (= number of windows).
Circuit above: all nodes n = 4, elements b = 5 → 5 − 4 + 1 = **2 meshes**; essential n = 2, essential branches b = 3 → 3 − 2 + 1 = **2** ✓, nodal = 2 − 1 = **1** equation.

Exam trap: Count with **one** definition throughout: all nodes with all elements, *or* essential nodes with essential branches. Mixing them gives a wrong mesh count. A circuit has 3 loops here but only 2 meshes.

2 Network Types & Turning Sources Off

Linear (R, L, C, linear dependent sources) vs non-linear (diode, transistor) · **bilateral** (R, L, C: same both ways) vs unilateral (diode) · **active** (has a source) vs passive. Superposition, Thevenin and Norton need a **linear** network; reciprocity also needs it bilateral.

To turn off …	replace by
ideal V-source	**short** (V = 0)
ideal I-source	**open** (I = 0)
practical source	keep its internal R
dependent source	never off

Memory trick: "Kill a voltage source with a **wire**, kill a current source with **air**." A dead V-source keeps 0 V across it (a wire); a dead I-source passes 0 A (a gap).

Which method? Pick the one with fewer equations

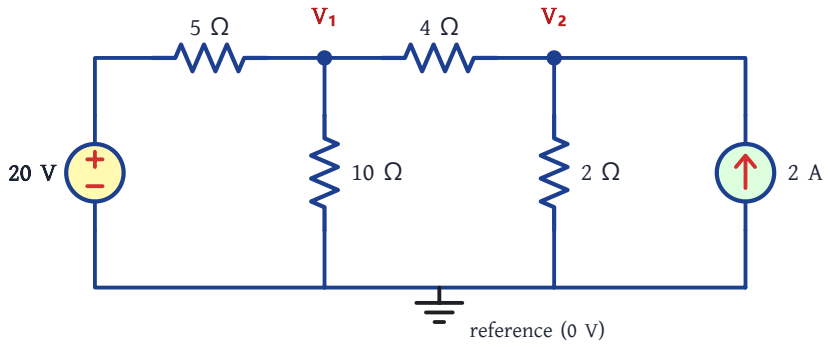
	Nodal analysis	Mesh analysis
Unknowns · law	node voltages · KCL	mesh currents · KVL
Equations	n − 1 (− 1 per V-source)	b − n + 1 (− 1 per I-source)
Easy with	**current sources**	**voltage sources**
Applies to	**any** circuit (planar or not)	**planar** circuits only

3 Nodal Analysis (Node-Voltage Method)

Idea: Unknowns are the **node voltages** measured from one **reference node** (ground). Write **KCL** at every other node, with each branch current written by Ohm's law as $(V_{this} - V_{other})/R$.

1. Reference = node with most branches
2. Name the rest V_1, V_2 … (n − 1 unknowns)
3. KCL: Σ out via R = Σ source current in
4. Solve (substitution or Cramer)
5. Branch currents from node voltages

Memory trick: "**My node minus my neighbour, over the resistor**": write every resistor current as leaving the node you stand on; the signs then take care of themselves.



Worked example ✍️: Find V_1, V_2 and every branch current.

1. KCL at V_1: $(V_1 - 20)/5 + V_1/10 + (V_1 - V_2)/4 = 0$ → ×20: $\mathbf{11V_1 - 5V_2 = 80}$
2. KCL at V_2: $(V_2 - V_1)/4 + V_2/2 = 2$ → ×4: $\mathbf{-V_1 + 3V_2 = 8}$
3. $V_1 = 3V_2 - 8$ → $11(3V_2 - 8) - 5V_2 = 80$ → $28V_2 = 168$ → $\mathbf{V_2 = 6\ V}$, $\mathbf{V_1 = 10\ V}$
4. Currents: 5 Ω: (20 − 10)/5 = 2 A · 10 Ω: 1 A · 4 Ω: 1 A · 2 Ω: 3 A. Check KCL at V_2: 1 + 2 = 3 ✓

By inspection: [G][V] = [I]:

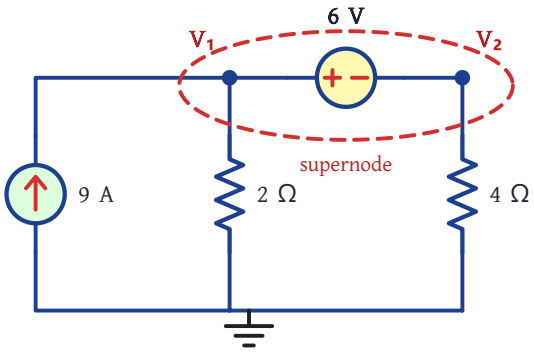
G_{kk} = Σ conductances at node k; G_{kj} = − (conductance between k and j); [I] = source currents in (20 V/5 Ω → 4 A):

$\mathbf{0.55V_1 - 0.25V_2 = 4}$ · $\mathbf{-0.25V_1 + 0.75V_2 = 2}$

Exam trap: A resistor **in series with a current source** never appears in the nodal equation: the source fixes that branch current. [G] is symmetric only when there are **no dependent sources**.

IT link: Circuit simulators (SPICE and its descendants) use **Modified Nodal Analysis**: they build this very [G] matrix for thousands of nodes and solve it numerically, like a linear-algebra library.

Supernode: a voltage source between two non-reference nodes



Enclose both nodes **and** the source. Write **one KCL for the whole supernode** plus the **constraint** from the source.

KCL: $\mathbf{V_1/2 + V_2/4 = 9}$

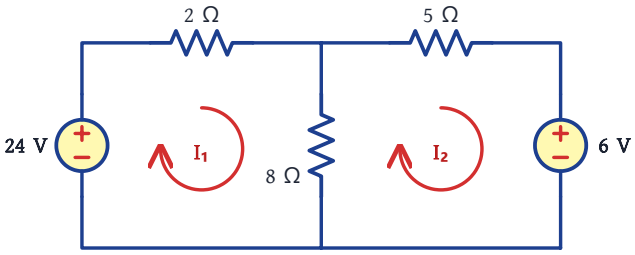
Constraint: $\mathbf{V_1 - V_2 = 6}$

→ $V_1/2 + (V_1 - 6)/4 = 9$ → $3V_1 = 42$ → $\mathbf{V_1 = 14\ V}$, $V_2 = 8$ V

Check: 7 A + 2 A = 9 A ✓. If the V-source touches the reference, that node voltage is simply **known**.

4 Mesh Analysis (Mesh-Current Method)

Idea: Give every window (mesh) a **mesh current**, all clockwise. Write **KVL** round each mesh. An element shared by two meshes carries the **difference** of their currents. Works for **planar** circuits only.



Mesh 1: $-24 + 2I_1 + 8(I_1 - I_2) = 0$

→ $\mathbf{10I_1 - 8I_2 = 24}$

Mesh 2: $8(I_2 - I_1) + 5I_2 + 6 = 0$

→ $\mathbf{-8I_1 + 13I_2 = -6}$

Cramer: $\Delta = 130 - 64 = 66$

$I_1 = 264/66 = \mathbf{4\ A}$ · $I_2 = 132/66 = \mathbf{2\ A}$

8 Ω carries $I_1 - I_2 = \mathbf{2\ A}$ downward.

Checks ✍️: Power: the 24 V source gives 24 × 4 = 96 W = 32 (2 Ω) + 32 (8 Ω) + 20 (5 Ω) + 12 W absorbed by the 6 V source (I_2 enters its + end: it is being charged) ✓. **Nodal on the same circuit:** one unknown, V = 8 × 2 = 16 V: (16 − 24)/2 + 16/8 + (16 − 6)/5 = −4 + 2 + 2 = 0 ✓, so **1 equation instead of 2.**

By inspection: [R][I] = [V]:

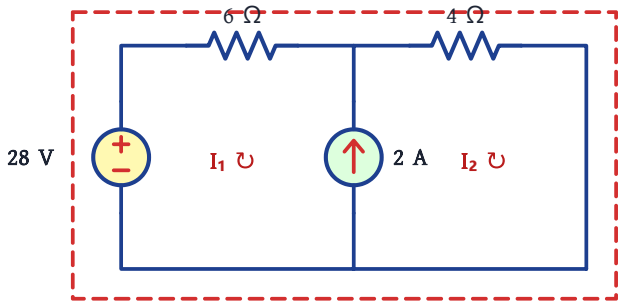
R_{kk} = total R round mesh k · R_{kj} = −(shared R)

V_k = source rises along the mesh current:

$10I_1 - 8I_2 = 24$ · $-8I_1 + 13I_2 = -6$

Exam trap: A mesh current is **not** always a branch current: in a shared branch the real current is $I_1 - I_2$. A resistor **in parallel with a voltage source** does not change any mesh current outside it.

Supermesh: a current source shared by two meshes



supermesh: go round the outside

KVL round the **outer path** (skip the current source): $6I_1 + 4I_2 = 28$

Constraint (2 A upward in the shared branch): $I_2 - I_1 = 2$

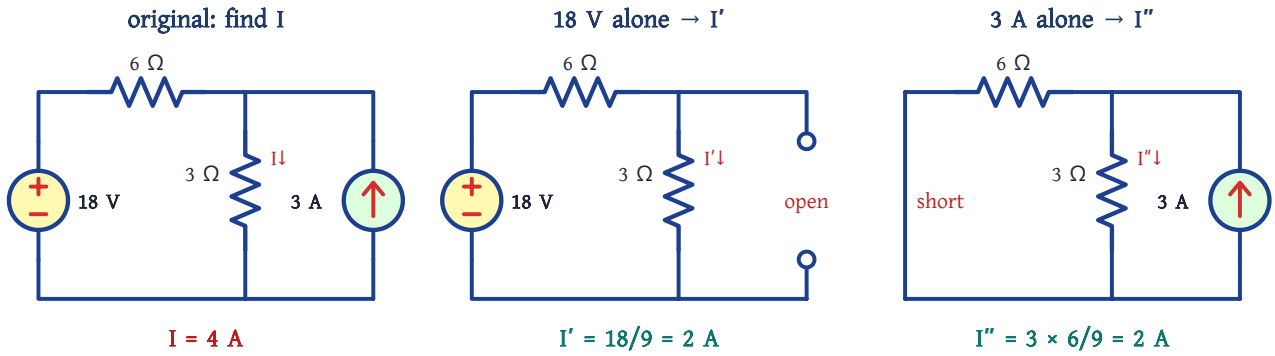
→ $10I_1 + 8 = 28$ → $I_1 = 2$ **A**, $I_2 = 4$ A

A current source on the **outer edge** of a mesh simply gives that mesh current.

Memory trick: duality: Nodal ↔ mesh: KCL ↔ KVL · V ↔ I · G ↔ R · **supernode** ↔ **supermesh** · current sources easy ↔ voltage sources easy · open ↔ short · Norton ↔ Thevenin.

5 Superposition Theorem

Statement: In a **linear** network with several independent sources, any branch current or voltage is the **algebraic sum** of the responses to each independent source acting **alone**, with all the other independent sources turned off.



one source at a time: V-source → short, I-source → open

Worked example ✍️: Current I in the 3 Ω resistor: 18 V alone → I′ = 18/(6 + 3) = 2 A; 3 A alone → current divider I″ = 3 × 6/(6 + 3) = 2 A; both downward, so **I = 4 A**. Nodal check: (V − 18)/6 + V/3 = 3 → V = 12 V → I = 12/3 = 4 A ✓.

Exam trap: **Power does not superpose:** $P(3\ \Omega) = 4^2 \times 3 = \mathbf{48\ W}$, but $2^2 \times 3 + 2^2 \times 3 = 24$ W ✗. Power $\propto I^2$ is not linear. Find the total current first, then the power.

Where it is used: a transistor amplifier is analysed twice, DC bias and AC signal separately (superposition); sources at different frequencies are handled one frequency at a time (lesson 3). Not valid for diodes or other non-linear elements.

Linearity = two properties:

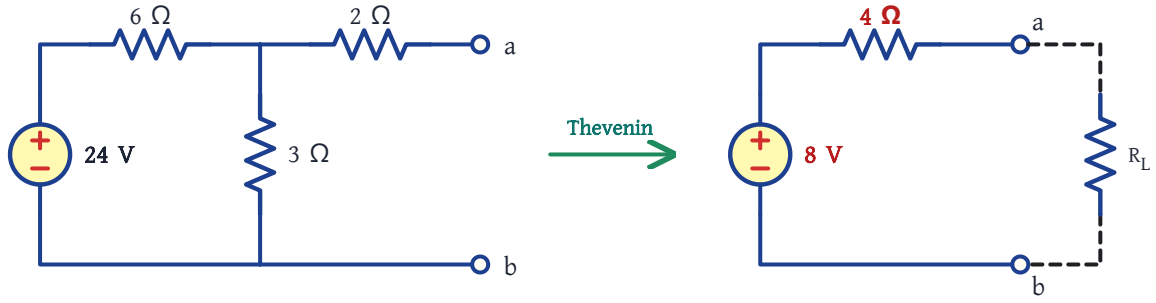
Homogeneity: k × input → k × output (18 V → 36 V makes I′ = 4 A, so I = 6 A)

Additivity: response to $(x_1 + x_2)$ = response to x_1 + response to x_2

Memory trick: "**One ON, rest OFF, add with signs.**" Number of sub-circuits = number of **independent** sources; dependent sources stay ON in every one.

6 Thevenin's Theorem

Statement: Any **linear two-terminal** network, seen from terminals a–b, can be replaced by a voltage source V_{th} **in series with** R_{th}. V_{th} = **open-circuit voltage** at a–b; R_{th} = resistance seen into a–b with all **independent** sources turned off. (Thévenin, **1883**; Helmholtz found it in 1853.)



whatever is inside, the load sees only V_{th} and R_{th}

1. Remove the load; mark a and b
2. $\mathbf{V_{th} = V_{oc}}$ across open a–b (any method: divider, nodal, mesh, superposition)
3. $\mathbf{R_{th}}$: independent sources off and look into a–b; or $R_{th} = V_{oc}/I_{sc}$; or apply a test source (needed with dependent sources)
4. Reconnect the load: $\mathbf{I_L = V_{th}/(R_{th} + R_L)}$

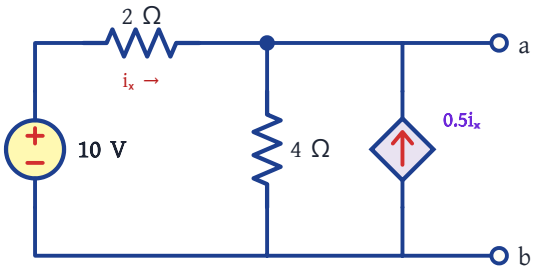
Worked example ✍️: Open a–b: no current in 2 Ω, so $V_{th} = 24 \times 3/(6 + 3) =$ **8 V**. Short the 24 V: $R_{th} = (6 \parallel 3) + 2 = 2 + 2 =$ **4 Ω**.

R_L	2 Ω	4 Ω	12 Ω
$I_L = 8/(4 + R_L)$	1.33 A	1 A	0.5 A
$P_L = I_L^2 R_L$	3.56 W	**4 W**	3 W

One equivalent, any number of loads: that is why examiners love Thevenin.

IT link: A battery, USB port or phone charger **is** a Thevenin source: EMF = V_{th}, internal resistance = R_{th}. Terminal voltage $V = V_{th} - I{\cdot}R_{th}$ sags as the load draws more current.

Thevenin with a dependent source (written favourite)



V_{oc}: KCL at a: $i_x + 0.5i_x = V/4$ with $i_x = (10 - V)/2$ → $0.75(10 - V) = 0.25V$ → **$V_{oc} = 7.5$ V**

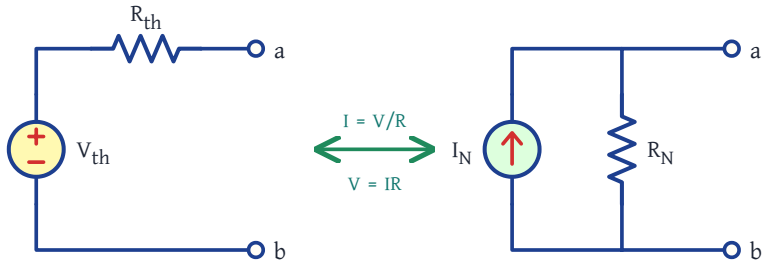
I_{sc}: a–b shorted, $V = 0$ → $i_x = 5$ A, 4 Ω carries 0 → $I_{sc} = 1.5 \times 5 =$ **7.5 A**

$R_{th} = V_{oc}/I_{sc} =$ **1 Ω**, not $2 \parallel 4 = 1.33$ Ω: the dependent source (a CCCS) changes R_{th}.

Exam trap: Never switch off a dependent source when finding R_{th}; with one, R_{th} may even be **negative**. The equivalent is exact **only at the terminals**, not for power lost inside.

7 Norton's Theorem

Statement: The same two-terminal network can be replaced by a current source **I_N in parallel with R_N**. I_N = **short-circuit current** through a–b; **$R_N = R_{th}$**. (E. L. Norton, Bell Labs, 1926.)

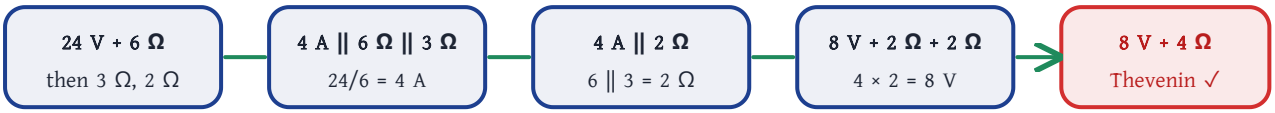


Memory trick: "TVS · NIP": Thevenin = Voltage in Series; Norton = current (I) in Parallel. The I_N arrow points to **a**, the + end of V_{th}.

Same network as p. 6: short a–b → $R = 6 + (3 \parallel 2) = 7.2$ Ω, source current $24/7.2 = 3.33$ A, divider into 2 Ω: $3.33 \times 3/5 =$ **$I_N = 2$ A** $= V_{th}/R_{th} = 8/4$ ✓.

8 Source Transformation as a Tool

Rule (lesson 1): V_s in series with R ⇄ $I_s = V_s/R$ in parallel with the **same** R. Used repeatedly, it collapses a ladder to its Thevenin form without any equations:



Exam trap: An **ideal** V-source (no series R) has no Norton form and an ideal I-source (no parallel R) has no Thevenin form. Do not transform the branch whose current you are asked for.

9 Maximum Power Transfer Theorem

Statement: A linear DC network delivers **maximum power** to a resistive load when $R_L = R_{th}$. Then $P_{max} = V_{th}^2/(4R_{th})$ and the efficiency is only **50%**.

Derivation ✍️:

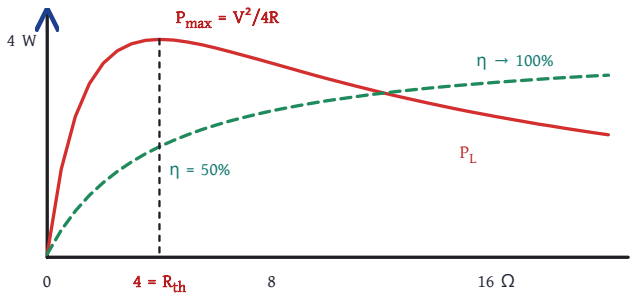
$P_L = I^2R_L = V_{th}^2R_L/(R_{th} + R_L)^2$

$dP_L/dR_L = V_{th}^2[(R_{th} + R_L)^2 - 2R_L(R_{th} + R_L)]/(R_{th} + R_L)^4 = 0$

$\rightarrow R_{th} + R_L = 2R_L \rightarrow \mathbf{R_L = R_{th}}$

$P_{max} = V_{th}^2R_{th}/(2R_{th})^2 = \mathbf{V_{th}^2/4R_{th}}$

$\eta = R_L/(R_L + R_{th}) = 1/2$



V_{th} = 8 V, R_{th} = 4 Ω

R_L (Ω)	1	2	4	8	16
P_L (W)	2.56	3.56	**4**	3.56	2.56
η (%)	20	33	**50**	67	80

Memory trick: R_L and R_{th}^2/R_L give the **same** power (1 Ω and 16 Ω, 2 Ω and 8 Ω). "Match for max, pay a 50% tax."

Exam trap: If R_L is fixed and the **source** resistance can be chosen, load power is maximum at $R_s = 0$, not $R_s = R_L$. AC: $Z_L = \mathbf{Z_{th}^*}$ **(conjugate)**, $P_{max} = |V_{th}|^2/4R_{th}$; a purely resistive AC load takes $R_L = |Z_{th}|$.

IT link: Matching is for **signals**, never for power grids (50% loss): 50 Ω RF coax and antennas, 75 Ω TV/CCTV coax, 100 Ω Ethernet twisted pair. It also stops **reflections** on the line.

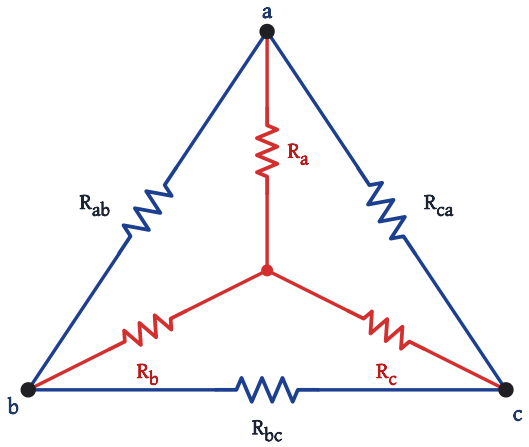
10 Reciprocity & Millman's Theorem

Reciprocity ✍️: In a **linear, bilateral** network with a **single** source, V at port 1 giving I in a short at port 2 means the same V at port 2 gives the **same** I at port 1 (fails with dependent sources, diodes, transistors). Check on a ladder 2 Ω (series) – 3 Ω (shunt) – 6 Ω (series). 12 V at port 1: $12/(2 + 3\|6) = 3$ A, into 6 Ω: $3 \times 3/9 =$ **1 A**. 12 V at port 2: $12/(6 + 3\|2) = 1.67$ A, into 2 Ω: $1.67 \times 3/5 =$ **1 A** ✓

Millman ✍️: Parallel branches (V_k in series with R_k): $\mathbf{V = \Sigma(V_k/R_k) \,/\, \Sigma(1/R_k)}$. 12 V/2 Ω, 6 V/3 Ω and a plain 6 Ω: $V = (6 + 2 + 0)/(½ + ⅓ + ⅙) =$ **8 V**, $R_{eq} = 1$ Ω.

11 Star–Delta (Y–Δ) Transformation

Idea: A **Δ (delta, π)** and a **Y (star, T)** of resistors are equivalent when the resistance between **every pair of terminals** is the same. Use it when resistors are neither in series nor in parallel (bridges, lattices).



blue = Δ, red = Y (drawn inside it)

Δ → Y:

$R_a = R_{ab}R_{ca} \,/\, (R_{ab} + R_{bc} + R_{ca})$

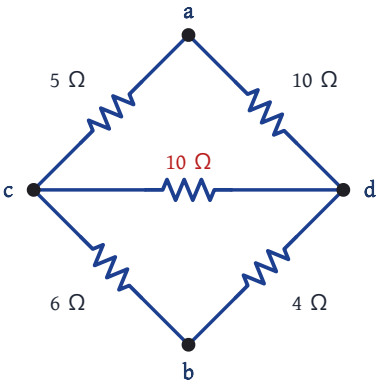
$R_b = R_{ab}R_{bc} \,/\, \Sigma_\Delta \cdot R_c = R_{bc}R_{ca} \,/\, \Sigma_\Delta$

Y → Δ: with $S = R_aR_b + R_bR_c + R_cR_a$:

$R_{ab} = S/R_c \cdot R_{bc} = S/R_a \cdot R_{ca} = S/R_b$

Balanced (all equal): $\mathbf{R_Y = R_\Delta/3}$, $R_\Delta = 3R_Y$

Memory trick: **Δ → Y:** "the two arms touching the node, multiplied, over the sum of all three" (looks like product/sum of parallel, so Y is **smaller**). **Y → Δ:** "sum of the pair-products over the **opposite** arm" (always bigger).



unbalanced bridge: 5/6 ≠ 10/4

Worked example ✍️: Find R_{ab}.

1. Δ(a, c, d): 5, 10, 10 → Σ = 25
2. $R_a = 5{\times}10/25 = 2$ · $R_c = 5{\times}10/25 = 2$ · $R_d = 10{\times}10/25 = 4$ Ω
3. Now series–parallel: $(2 + 6) \,\|\, (4 + 4) = 8 \,\|\, 8 = 4$ Ω
4. $R_{ab} = 2 + 4 =$ **6 Ω**

Balanced Wheatstone bridge: If P/Q = R/S, the middle (galvanometer) arm carries **no current**: drop it. 10, 20 | 30, 60 → R_{ab} = 30 || 90 = **22.5 Ω**, whatever the middle resistor.

Cube of 12 equal R	R_{eq}
across a body diagonal	**5R/6**
across a face diagonal	**3R/4**
across one edge	**7R/12**

Exam trap: Check for a **balanced bridge first** (then no Y–Δ work). Each Y resistor is **smaller** than both Δ resistors next to it; a bigger value means a wrong divisor.

Night-Before-the-Exam Sheet

Fact sheet:

Nodal: n − 1 equations (KCL)

Mesh: b − n + 1 equations (KVL)

$[G][V] = [I] \cdot [R][I] = [V]$

Branch current = $(V_{this} - V_{other})/R$

Supernode: 1 KCL + $V_1 - V_2 = V_s$

Supermesh: outer KVL + $I_2 - I_1 = I_s$

Off: V-source → short, I-source → open

Dependent sources: never off

$V_{th} = V_{oc} \cdot I_N = I_{sc}$

$R_{th} = R_N = V_{oc}/I_{sc}$

$I_L = V_{th}/(R_{th} + R_L)$

$P_{max} = V_{th}^2/4R_{th}$ at $R_L = R_{th}$, η = 50%

AC: $Z_L = Z_{th}^*$ for max power

Millman: $V = \Sigma(V_k/R_k)/\Sigma(1/R_k)$

Δ→Y: $R_a = R_{ab}R_{ca}/\Sigma_\Delta$

Y→Δ: $R_{ab} = \Sigma R_i R_j/R_c$

Balanced: $R_Y = R_\Delta/3$

Bridge balanced: P/Q = R/S

Top 14 one-liners for MCQs

1. Nodal analysis is based on KCL; mesh on KVL
2. Mesh analysis: planar circuits only
3. Superposition needs a linear network
4. Superposition can't give power
5. Thevenin: V-source in series with R
6. Norton: I-source in parallel with R
7. Thevenin 10 V, 5 Ω = Norton 2 A, 5 Ω
8. Max power at $R_L = R_{th}$; efficiency 50%
9. 12 V, 4 Ω network → P_{max} = 9 W
10. Reciprocity: linear + bilateral, one source
11. Three 9 Ω in Δ = three 3 Ω in Y
12. Cube of R, body diagonal: 5R/6
13. Ideal V-source: r = 0; ideal I-source: r = ∞
14. Tellegen: $\Sigma\, v_k i_k = 0$ (power balance)

Top traps: power never superposes · dependent sources stay on · resistor in series with an I-source is ignored in nodal · count nodes and branches with one definition · fixed R_L, variable R_s → $R_s = 0$ for max power · check for a balanced bridge before Y–Δ · ideal sources cannot be transformed

All tricks together: my node minus my neighbour · kill V with a wire, kill I with air · nodal ↔ mesh duality · one ON, rest OFF, add with signs · TVS · NIP · match for max, pay a 50% tax · R_L and R_{th}^2/R_L give equal power · Δ→Y product over sum, Y→Δ pair-products over the opposite arm



the three-step routine behind Thevenin, Norton and max power questions

✓ Practice MCQs (exam pattern, 10 marks)

1. Nodal analysis of a circuit with n nodes needs:
(a) n equations (b) b − n + 1
(c) n − 1 (d) n + 1

2. Superposition cannot directly give:
(a) power (b) current
(c) voltage (d) none

3. Norton's I_N is the ___ current at a–b:
(a) open-circuit (b) short-circuit
(c) load (d) leakage

4. Efficiency at maximum power transfer:
(a) 100% (b) 75%
(c) 25% (d) 50%

5. Three 9 Ω in delta equal a star of:
(a) 27 Ω each (b) 3 Ω each
(c) 9 Ω each (d) 4.5 Ω each

6. A supermesh forms when two meshes share a:
(a) current source (b) voltage source
(c) resistor (d) node

7. V_{th} 20 V, R_{th} 5 Ω, R_L 15 Ω → I_L:
(a) 4 A (b) 1.33 A
(c) 1 A (d) 2 A

8. Reciprocity fails in a network with:
(a) only R (b) a CCCS
(c) L and R (d) C and R

9. 10 A in parallel with 2 Ω = a source of ___ in series with 2 Ω:
(a) 5 V (b) 10 V
(c) 20 V (d) 40 V

10. 12 V source, 2 Ω internal resistance: maximum load power?
(a) 72 W (b) 36 W
(c) 9 W (d) 18 W

Answer key → 1-c · 2-a · 3-b · 4-d · 5-b · 6-a · 7-c · 8-b · 9-c · 10-d

Written question: A 30 V source in series with 6 Ω feeds node a; 3 Ω joins a to b (ground); a 2 A source injects current into a. Find the Thevenin equivalent at a–b, the load for maximum power and that power. 8 marks

1. **V_{th} by superposition:** 30 V alone: 30 × 3/9 = 10 V; 2 A alone: 2 × (6 ‖ 3) = 4 V → **V_{th} = 14 V** 3
2. **R_{th}:** 30 V → short, 2 A → open: 6 ‖ 3 = **2 Ω**. Check: I_{sc} = 30/6 + 2 = 7 A = 14/2 ✓ 2
3. Maximum power at **R_L = 2 Ω** 1; $P_{max} = 14^2/(4 \times 2)$ = **24.5 W** (I_L = 3.5 A) 2

Before the exam ✓

✓ Can count n − 1 and b − n + 1 correctly

Also practise (written):
① Prove the max power theorem
② Mesh analysis with a supermesh

- ✓ Can write nodal/mesh equations by inspection
- ✓ Can handle a supernode and a supermesh
- ✓ Know P_{max}, $\eta = 50\%$ and both Y–Δ formulas

③ Norton of the p. 6 network
④ R_{ab} of an unbalanced bridge